COMP2054 Tutorial Session 6: Heaps, BSTs, and BST Rotations

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Session outcomes

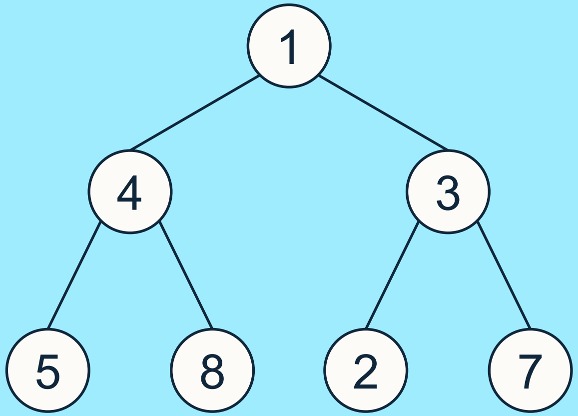
- Understand difference between heaps and BSTs.
- Know how to apply the various operations on heaps and BSTs.
- Be able to balance BSTs using rotations.

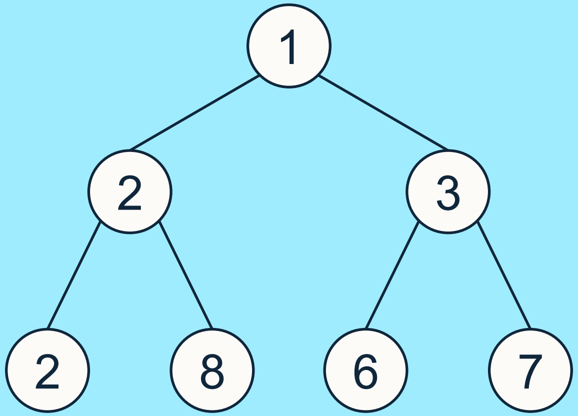
Trees

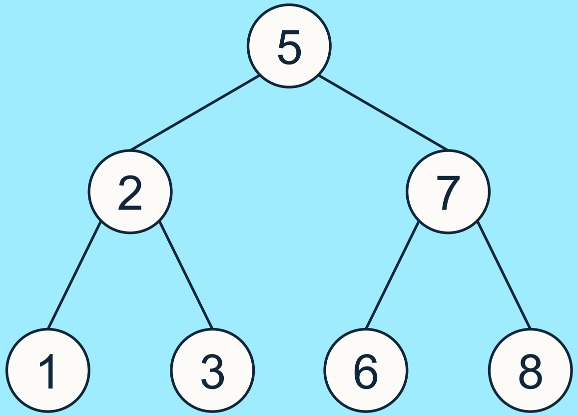
Heapsand BSTs

Quiz – Q1

Classify each of the following trees as a Heap, a BST, or neither:

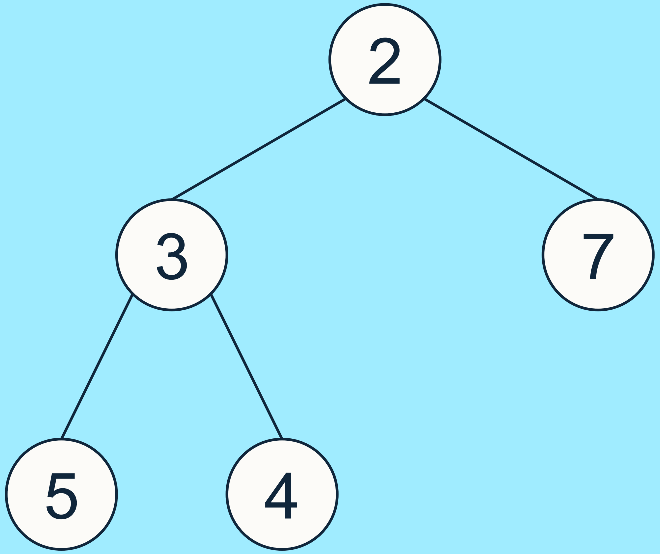






Quiz – Q2

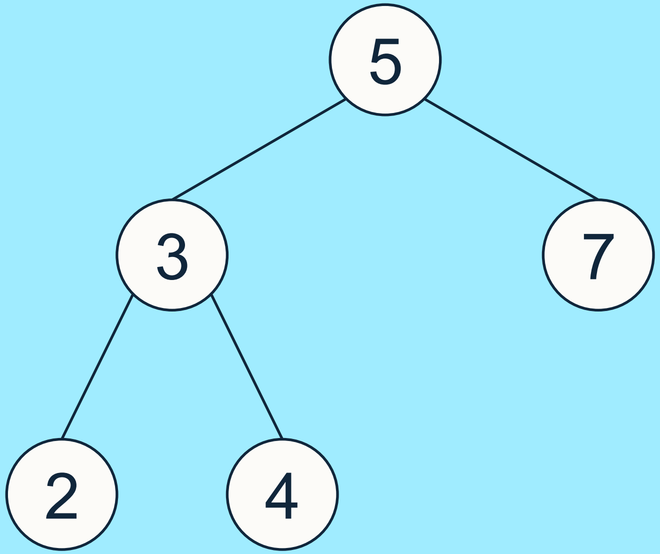
Which two nodes in the following heap should be swapped to create a BST?



Array Representation of Binary Trees

How can we represent the below BST in an array?

Index	[0]	[1]	[2]	[3]	[4]	[5]
Value						



Remember:

```
index(root) = 1
index(left_child, i) = 2i
index(right_child, i) = 2i + 1
```

Quiz – Q3

Given the two binary trees in an array-based format, identify which is a heap and which is a BST, and explain why.

Binary tree 1:

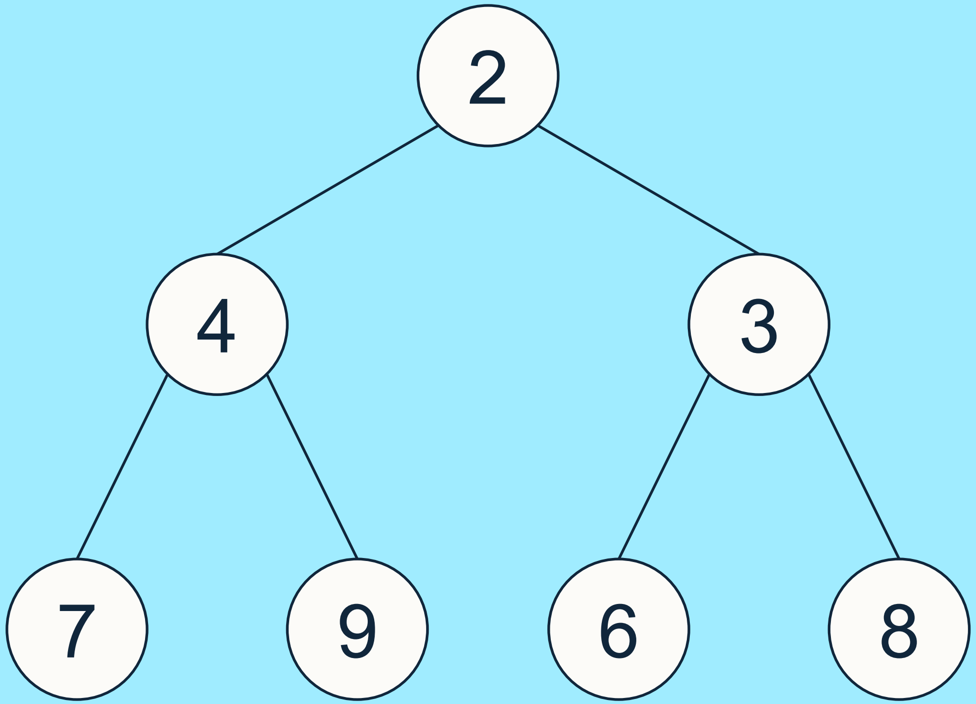
Index	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Value		2	3	5	4	6		

Binary tree 2:

Index	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Value		4	3	5	2			6

Operations on Heaps – insertItem(x)

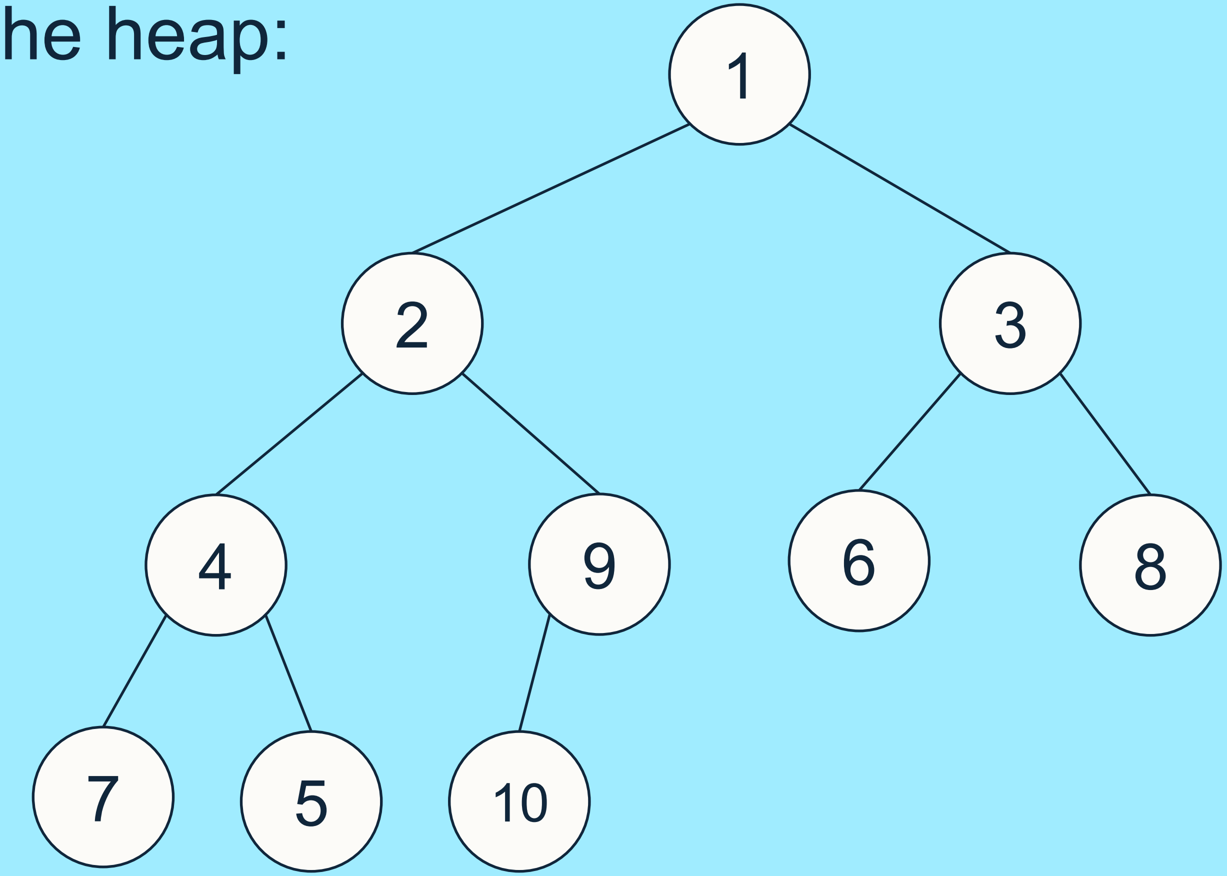
- Starting with the heap:



Perform insertItem(5), insertItem(1), then insertItem(10).

Operations on Heaps – removeMin()

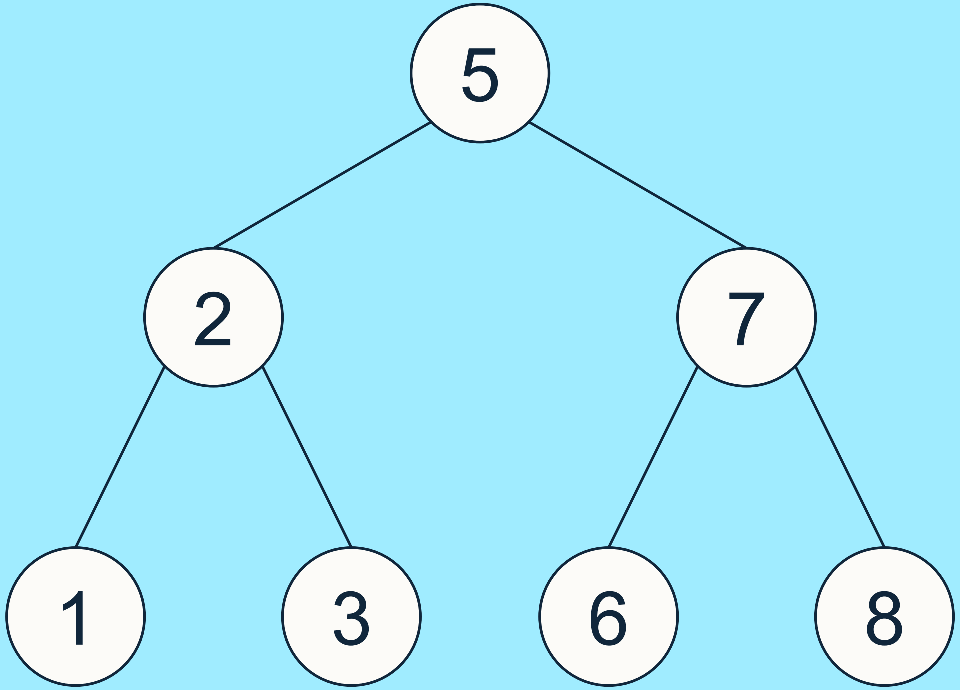
- Starting with the heap:



Perform removeMin(), swapping with smallest child (if less than current).

Operations on BSTs – insert(x)

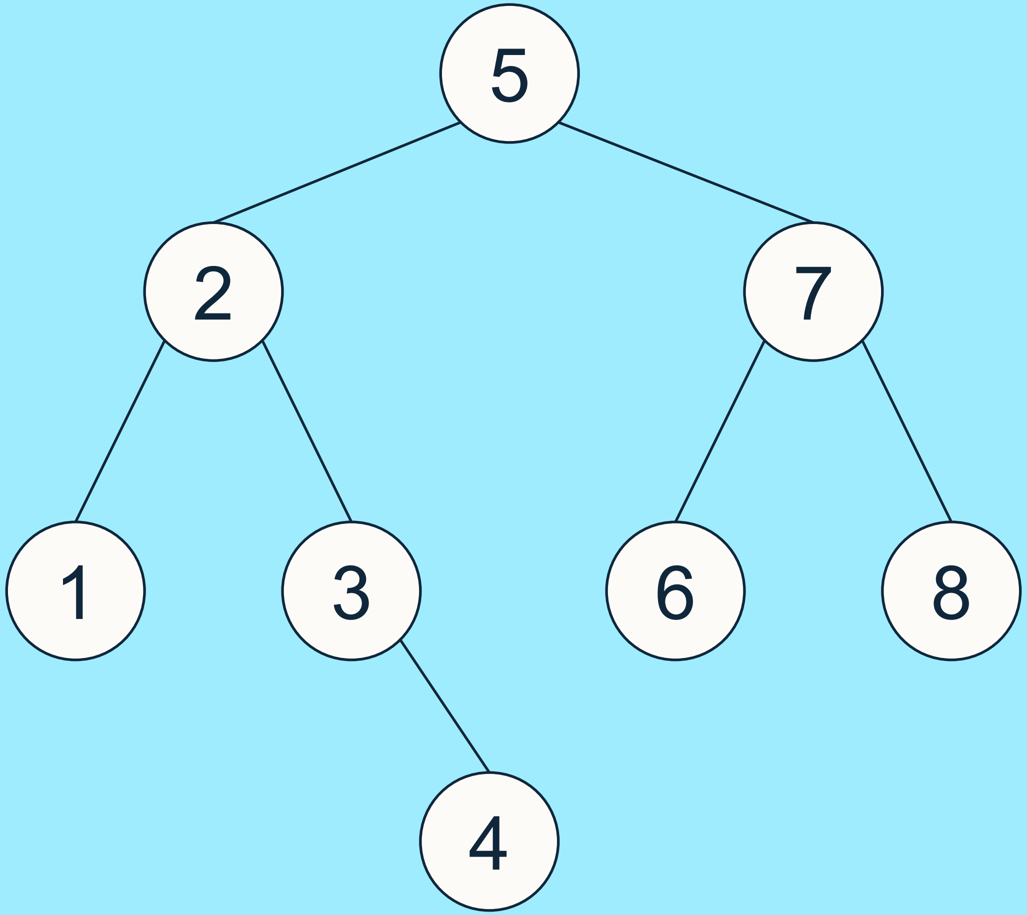
- Starting with the BST:



Perform insert(4).

Operations on BSTs – remove(x)

- Starting with the BST:



Perform remove(2), remove(7), then remove(5).

Questions: Heaps and BSTs

Question 1:

a. Draw the following as binary trees and label each with "Heap", "BST", or "Neither": $T_1 = [-, 1,2,4,7,8,5,6]$ and $T_2 = [-, 1,2,5,7,8,4,6]$.
b. If you identify either of the above as a **heap**, then perform `x = removeMin()` followed by `insertItem(x)`. Is the resulting heap identical?

Question 2: Use heapsort to sort the values [2,3,1,4,3',2',5]. Using your resulting sorted list, explain if heapsort is stable or not.

Question 3: With the following BST $[-, 6,2,10,1,5,7,11, -, -, 3, -, -, 9, -, -]$ perform at least the following operations and state the array-based representation after each step:

- Remove(7); remove(2); insert(7); remove(11); remove(10).

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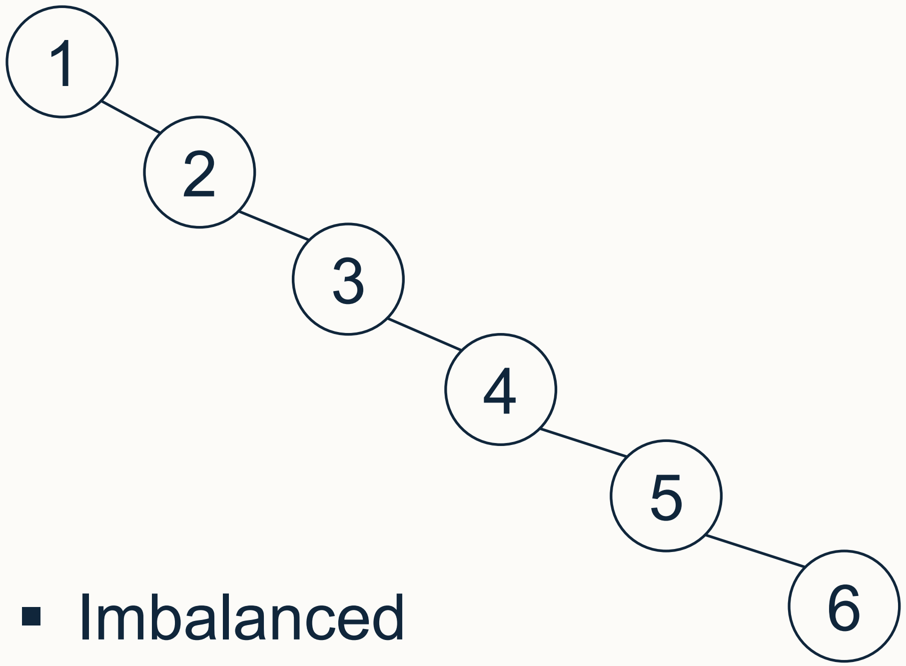
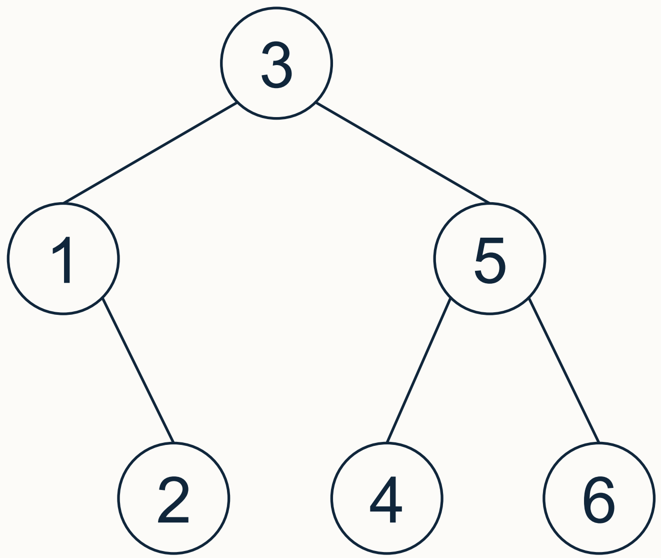
Rotations with BSTs

Motivation

- Important BST algorithms are O(height)
- For example, for a tree with $n = 6$ nodes:



- Imbalanced
- height = 5
- Operations will be O(n)



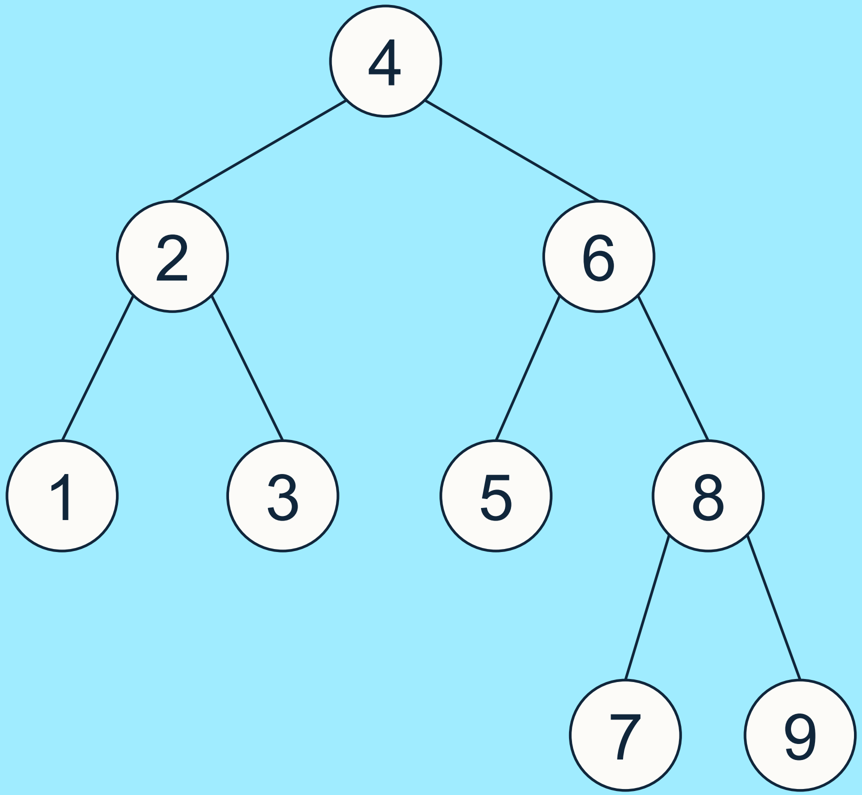
- Balanced
- height = 2
- Operations will be O(log n)

BST Rotations

- **Aim:** rebuild part of the tree in O(log n) time, so that the balance is maintained following insertion/deletion operations
- Resulting tree must have the same in-order traversal

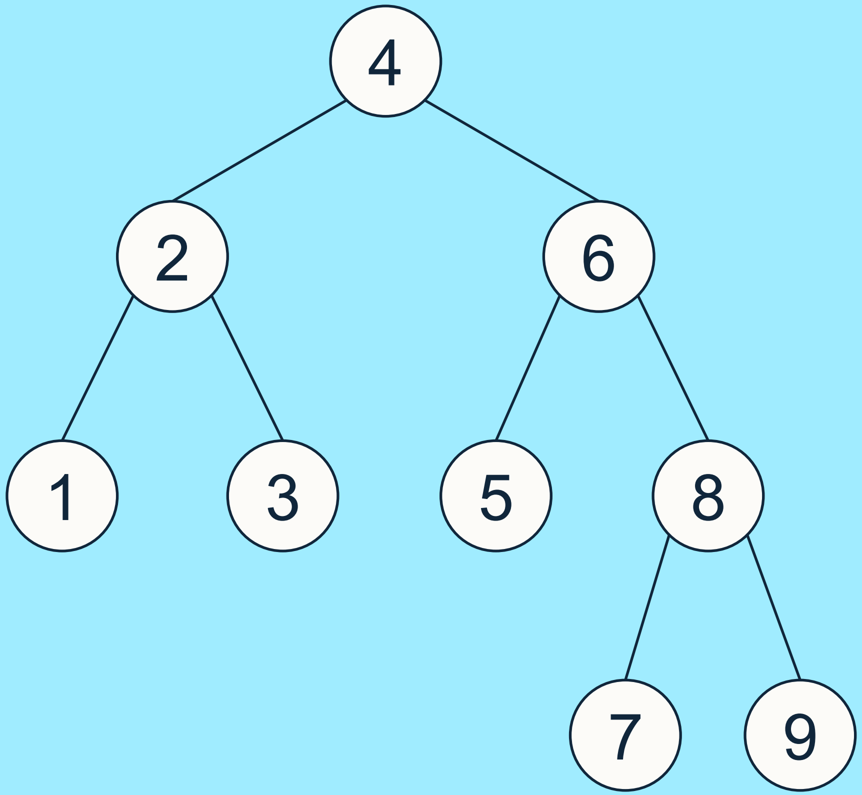
Single Rotation Example 1

Rotate with edge 2-4



Single Rotation Example 2

Rotate with edge 6-8

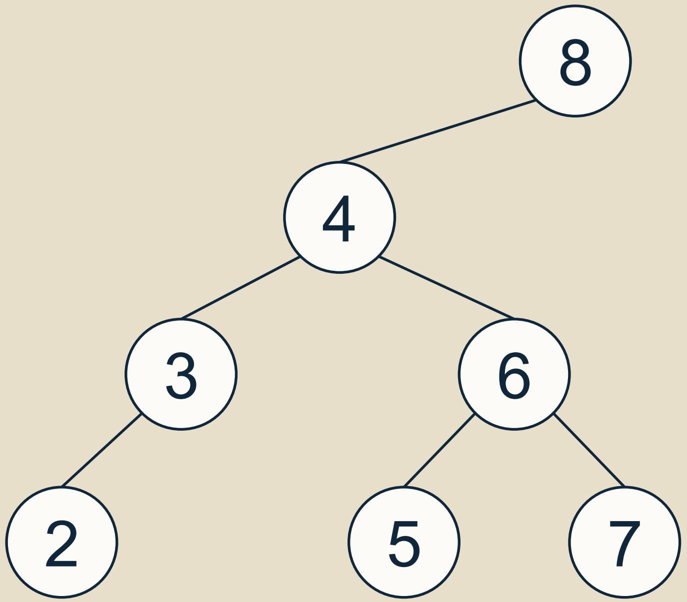


Questions

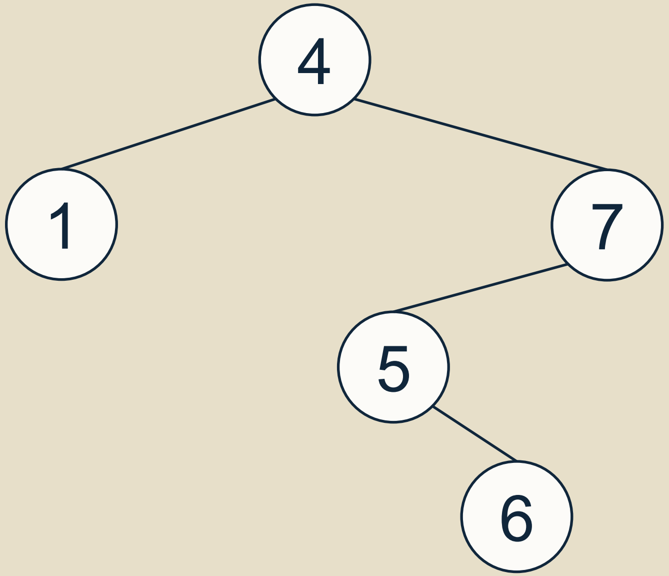
Q1. Starting from the BST to the right:

a. Insert 1

b. Perform a **single rotation** to reduce the height of the resulting tree



Q2. Given the BST to the right, find the **two pairs** of single rotations that **minimise** the height of the tree. Perform both pairs of rotations.



Thank you